Oct. 23

Test 2

Curve: 72-82 A, 62-71 B, 52-61 C, 41-51 D

- Option to re-do problems and having scores averaged.
- cannot decrease score on a problem.
- may use notes, textbook~~s~~, and me as resources. No other human or web-based material allowed.

- due Mon. Nov. 14 (1.5 weeks, 2 weekends.

Fourier Series

If $f(x)$ is defined over $-\pi \le x \le \pi$

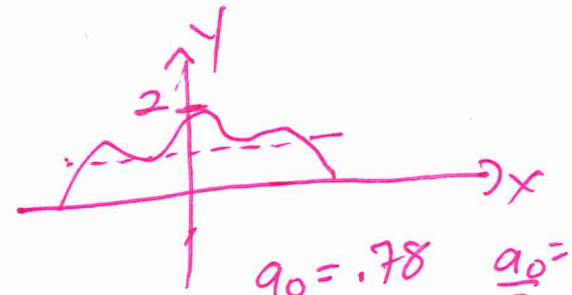
$$\text{F.S. of } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \sin(nx)\right]$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$

$\frac{a_0}{2}$ = average of the f'n.



$\frac{a_0}{2} = .78$ $\quad \frac{a_0}{2} = -.12$

If $f(x)$ is defined over $\boxed{-L \le x \le L}$

$$\text{F.S. of } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi}{L}x\right)\right]$$

$$\cancel{a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right) + b_n\,dx}$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

When does the F-S of $f(x)$ converge to $f(x)$ whenever $f(x)$ is C^1 with finite # of finite jumps on $-L \le x \le L$ or piecewise smooth

If $f(x)$ is defined over $\boxed{0 \le x \le L}$, Then we have 2 options:

① Extend The f'n evenly

$\Rightarrow$ cosine series

$$\text{F.S. of } f_{even}(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

$$= \frac{f(x^+) + f(x^-)}{2}$$ for all x, $f(x)$ is even extension repeated periodically.

② Extend The f'n oddly

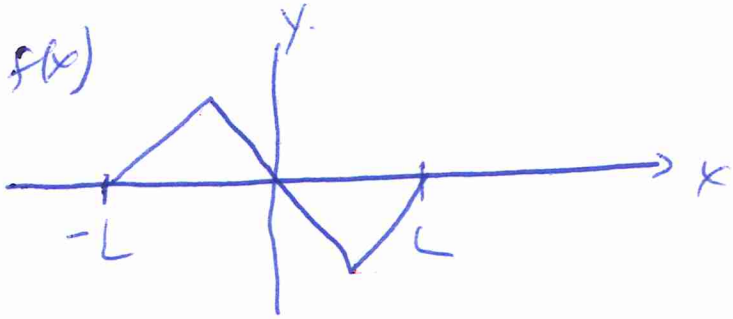
$\Rightarrow$ sine series

$$\text{F.S. of } f_{odd}(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) = \frac{f(x^+) + f(x^-)}{2}$$

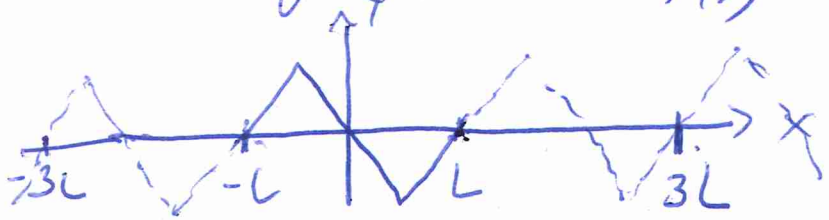
Note That for ① $\frac{a_0}{2}$ = avg. of f'n when extended evenly

② $f_{odd}(x=0) =$

ex) f(x)



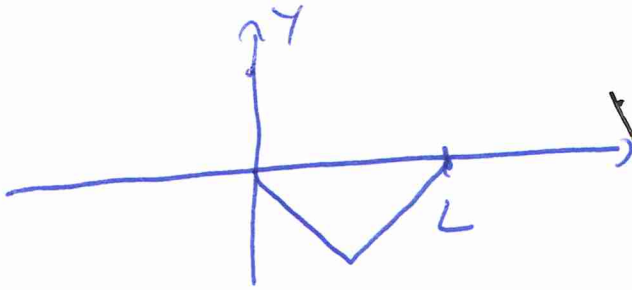
Graph FS of f(x)



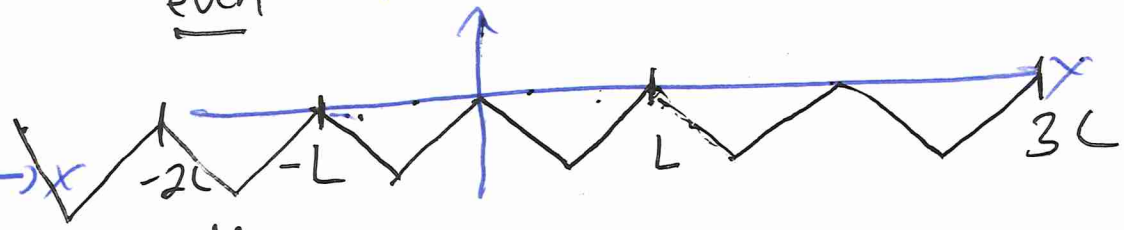
graph at least 3 periods

F.S. odd / evenly extended

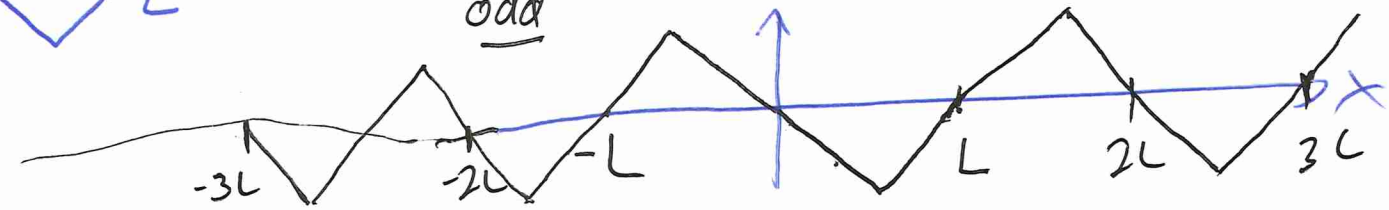
ex)



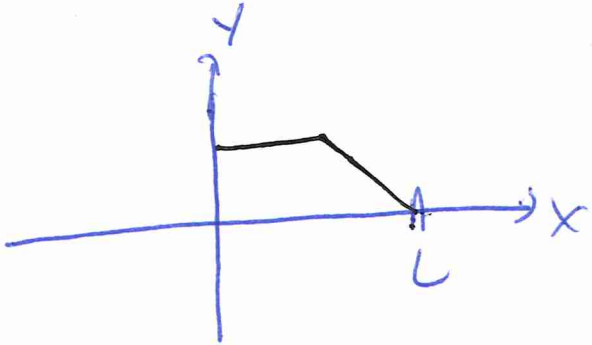
even



odd



ex)



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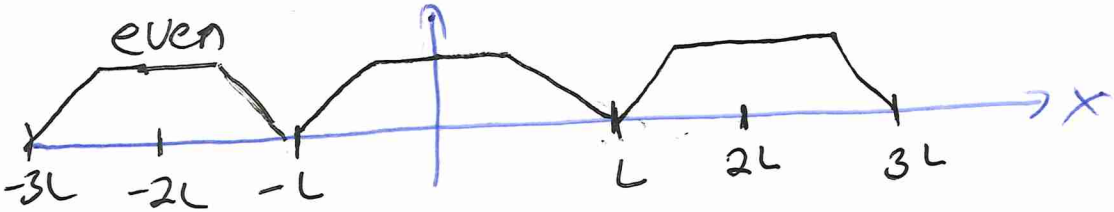

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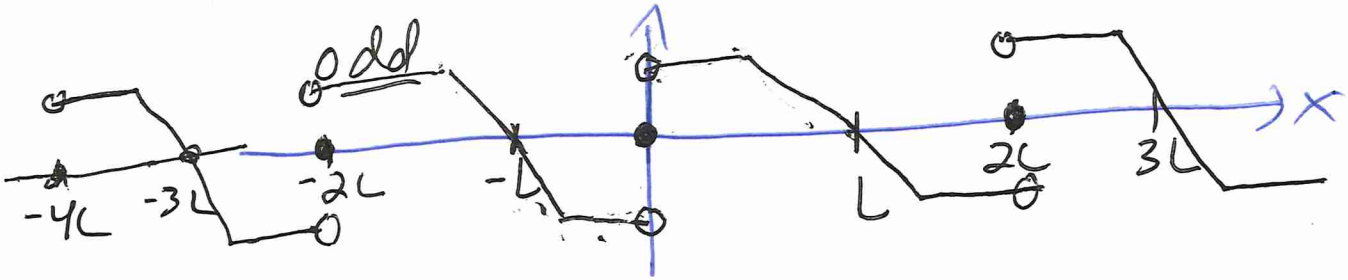

Ch 7 §7 sol'n to Dirichlet Problem on a unit circle ($\Delta u = 0$)

Dirichlet Problem

Assume ① Ω is regular (open, connect, bounded, Lipschitz bdry)

② $f \in C^0(\partial\Omega)$

Find $u \in C^0(\bar{\Omega}) \cap C^2(\Omega)$ so that

$$\Delta u = 0 \quad \Omega$$
$$u = f \quad \text{on } \partial\Omega$$

$\Omega = \{x, y \text{ in } \mathbb{R}^2 : x^2 + y^2 < 1\}$

$\Omega = \{(r, \theta) \text{ in } \mathbb{R}^2 : r < 1\}$

$\partial\Omega = \{(r, \theta) \text{ in } \mathbb{R}^2 : r = 1\}$

We know $r^n \cos(n\theta)$, $r^n \sin(n\theta)$ are harmonic

+ we know $\{1, \cos(n\theta), \sin(n\theta)\}$ for a basis for piecewise smooth f'ns.

Hypothesize

$$u(r,\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left[A_n r^n \cos(n\theta) + B_n r^n \sin(n\theta)\right]$$

$$0 \le r < 1$$

Not responsible for

is harmonic.

$$\Delta\left[\frac{A_0}{2} + \sum_{n=1}^{\infty}\left[\qquad\right]\right]$$

$$= \Delta\left(\frac{A_0}{2}\right) + \sum_{n=1}^{\infty}\left[A_n \underbrace{\Delta(r^n\cos(n\theta))}_{0} + B_n \underbrace{\Delta(r^n\sin(n\theta))}_{0}\right]$$

$$= 0$$

$$= 0$$

& by Thm If $\Delta u = 0$ & $u \in C^0 \Rightarrow u \in C^\infty$

$\Rightarrow u \in C^2(\Omega)$

We still need to show that

$$u(1,\theta) = f(\theta)$$

At $r=1$ we need

$$f(\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \left[A_n \cos(n\theta) + B_n \sin(n\theta)\right]$$

This is the F.S. of $f(\theta)$!

replace $\frac{A_0}{2} \to \frac{a_0}{2}$ $\quad A_n \to a_n \quad B_n \to b_n$

a_0, a_n, b_n are FS of $f(\theta)$

On unit disk $= \Omega$ $\quad 0 \le r < 1$

$\Delta u = 0 \quad \Omega \quad -\pi \le \theta \le \pi$

$u = f \quad \partial\Omega$

① Find the F.S of f

② $u(r,\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n r^n \cos(n\theta) + b_n r^n \sin(n\theta)\right]$

don't forget

One last thing.

Does $\lim_{r \to 1} u(r,\theta) = f(\theta)$ i.e. is $u \in C^0(\bar{\Omega})$

~~We need~~

We can show this directly, but we need an integral representation of $u(r,\theta)$ to show this.